

TEST REPORT: TR-BMS-2026-8801

EUT: Automotive Battery Control Unit BCU-800V | HW Rev 3.0

Standards: ISO 26262-5:2018, UN ECE R100 Rev 3, ISO 6469-3

1. Executive Test Summary

This official laboratory test report certifies electrical compliance and safety testing of the BCU-800V. Testing was conducted under ISO/IEC 17025 accredited laboratory conditions across input voltage, current sensing, isolation breakdown, sleep power, and cell monitoring precision.

2. Electrical Operating Range and Measurement Precision

Pack Voltage Sweep (REQ-BCU-001):

The BCU was tested across 400.0 V, 600.0 V and 800.0 V DC input levels with full telemetry accuracy. Zero overvoltage latching or false alarms recorded across 50 power cycling iterations.

Current Shunt Sensor Accuracy (REQ-BCU-002):

Current sensor measured 350.0 A continuous discharge with 0.18% error across 4-hour soak test. Thermal rise at shunt terminals was 18.2 °C above ambient.

Cell Voltage Measurement Precision (REQ-BCU-006):

Cell monitoring ASIC calibrated across 2.0V to 4.5V with maximum recorded error of +/- 0.85 mV. Total cell sensing channel dispersion satisfied the +/- 1.5 mV tolerance requirement.

Quiescent Sleep Current (REQ-BCU-005):

Deep sleep quiescent current measured at 112.4 uA at 12.0 V auxiliary supply, well below 150.0 uA limit.

3. Dielectric Isolation, Pre-Charge & Telemetry Verification

Dielectric Withstand Testing (REQ-BCU-003):

Dielectric withstand test: 2,500 V AC applied for 60.0 s, measured leakage current 0.28 mA, verdict PASS. Test conducted between HV pack bus and low-voltage 12V harness.

Pre-Charge Circuit Timing (REQ-BCU-009):

DC bus pre-charge timing test: 95% voltage reached in 142.0 ms with pre-charge resistor peak temperature within bounds.

CAN-FD Telemetry Verification (REQ-BCU-004):

CAN-FD bus evaluated at 5.0 Mbps across 10,000,000 frames with zero frame errors recorded under 90% bus utilization.

Overcurrent Verification Note (REQ-BCU-015):

Single-point overcurrent verification conducted at 200.0 A setting only; multi-point curve characterization incomplete.